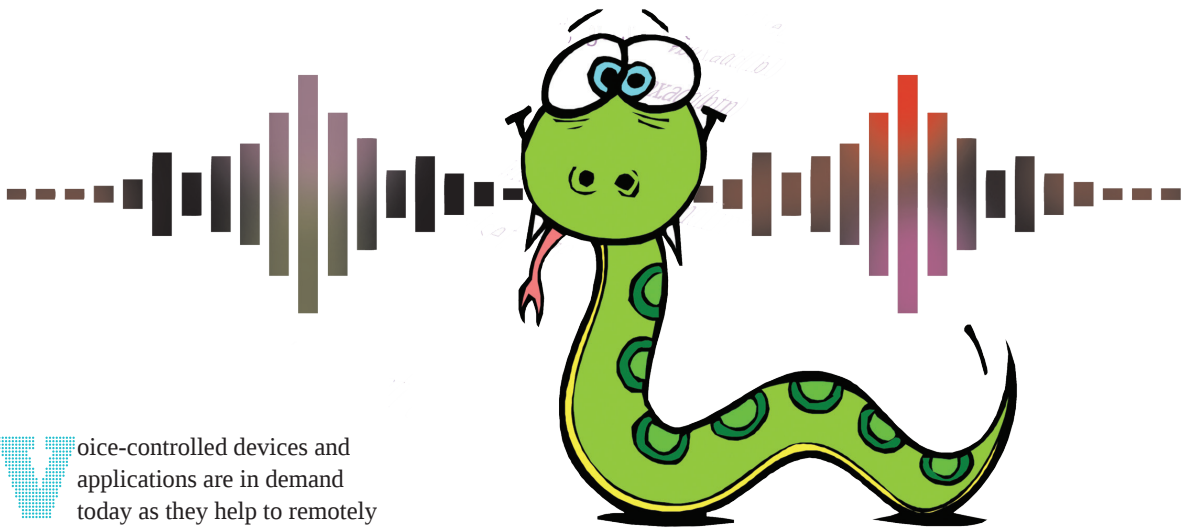


How Python Can Help to Process Audio Waves

Python is one of the most versatile and capable computer programming languages in use today. This article dives into the world of audio waves and plays with them using Python.



Voice-controlled devices and applications are in demand today as they help to remotely access smart home devices, give navigation commands to robots and autonomous vehicles, and so on. Audio signal processing is booming, be it for industrial applications, health care, the financial sector, etc. In this article, we are going to get an insight into the use of Python tools for audio processing. We begin with an overview of this field and then move to the implementation aspects.

An audio signal can be categorised into: (i) speech uttered by humans (also referred to as voice), (ii) music generated by an instrument, (iii) other sounds such as a dog barking, birds chirping, surrounding noise including environmental noise, machinery sounds, etc (Figure 1). In this article, we restrict

ourselves to speech and music signals; however, the processing methodology of all such files remains the same.

Speech perception in humans

Speech perception refers to the hearing ability of humans, and is mainly done through the ears. However, the structure of the human ear is complex. The outer ear is what we see. The middle ear is like a path for sound to travel towards the inner ear. The inner ear contains the most sensitive part of the human perception system, known as the cochlea. It is a snail-like structure that contains basilar membranes which act like

frequency receptors. The hair cells in these membranes activate at specific frequencies, and are arranged in a logarithmic scale such that they have higher resolution at lower frequencies and lower resolution at higher frequencies. Due to this arrangement, humans find it difficult to distinguish between two different frequency tones in the higher ranges. For example, it is easy to perceive the tonal difference between two sounds having frequencies 100Hz and 200Hz, which are 100Hz apart. However, it becomes almost impossible for humans to distinguish between two sounds with frequencies 10kHz and